DAY 21 | PRE-READ

How AI Writes

A light read before Day 21. About 15 minutes, no code required.

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Why read this?

The final weeks of the program are about generative AI: tools like ChatGPT, Claude and Gemini, and how to build reliable products with them. Tomorrow you learn how these models work and how to give them instructions that get good results.

1. A very well-read autocomplete

Your phone suggests the next word as you type. A large language model (LLM) does the same thing, but it has learned from an enormous amount of text, so its suggestions can form whole essays, emails or programs. It writes one small piece at a time, called a **token**, each time choosing what is most likely to come next.

2. Paying attention

In "I sat on the bank of the river", you know "bank" means the riverside, not a place for money, because you noticed the word "river". Modern models do this with a mechanism called **attention**: every word looks at the other words to work out what it means in this sentence.

3. Experts on call

A hospital does not send every patient to every doctor. A receptionist sends each patient to the right specialists. Some large models work the same way, with many specialist sub-networks and a router that picks a few for each piece of text. This is called **mixture of experts**.

4. Good instructions, good results

If you ask a new colleague to "write something about sales", you get something vague. If you say who it is for, what to include, how long, and show an example, you get what you need. Instructions to AI are called **prompts**, and writing them well is called **prompt engineering**.

Prompt part	Example
Role	You are a friendly support agent.
Task	Reply to this customer complaint.
Context	The order was delayed by two days due to weather.
Format	Three short sentences, no jargon.
Example	A sample of a great reply.

Key idea: AI can sound confident and still be wrong. Always check important facts.

5. About tomorrow's assessment

Tomorrow includes an MCQ assessment and a milestone check on your project. Have your full stack app and repository ready to show.

6. Think about these before class

- Ask an AI assistant the same question twice. Are the answers identical? Why might they differ?
- Write the best instruction you can for a task you do weekly.
- Where would a confident but wrong answer be dangerous in your work?

Kotoba (言葉): words, language